

# Generative AI and LLM Foundations Solutions

## Instructor guide and marking framework for PU B04 C01

This guide provides expected reasoning, common errors and a consistent scoring framework. Accept technically valid alternatives when the learner states assumptions and supports the conclusion.

### General marking framework

| Score | Standard                                                                                 |
|-------|------------------------------------------------------------------------------------------|
| 4     | Correct conclusion, valid reasoning, appropriate evidence and operational interpretation |
| 3     | Substantially correct with a minor omission                                              |
| 2     | Partial structure or correct idea without adequate reasoning                             |
| 1     | Relevant attempt with major gaps                                                         |
| 0     | No relevant evidence                                                                     |

### Exercise 1 Concept distinctions

Explain the difference among a discriminative model, a generative model and a retrieval system. Give one institutional use for each.

**Expected answer:** A discriminative model estimates an outcome conditional on features; a generative model learns a distribution from which outputs can be produced; retrieval selects external records. Suitable examples are classification, drafting and evidence lookup. Do not describe retrieval as training.

**Common error:** Conflating fluent language with verified truth.

Marking: award up to 4 points using the general framework.

### Exercise 2 Autoregressive generation

Starting from the sequence-probability factorization, explain what a language model estimates at each generation step and why high probability does not establish truth.

**Expected answer:** The model estimates  $p(x_t | x_1, \dots, x_{t-1})$  and repeatedly selects a next token. Sequence likelihood measures compatibility with model and context, not agreement with verified reality.

**Common error:** Reporting a notebook result without interpreting its boundary.

Marking: award up to 4 points using the general framework.

## Exercise 3 Tokenization

Tokenize one English and one French sentence using the notebook. Compare token counts, unknown-token behaviour and implications for multilingual service delivery.

**Expected answer:** The learner should run both sentences, report counts and explain that tokenizer vocabulary and segmentation affect context use, cost and language equity. Unknown-token results must be interpreted relative to the demonstration tokenizer.

**Common error:** Treating a prompt as the only control.

Marking: award up to 4 points using the general framework.

## Exercise 4 Context boundary

Design a context budget for summarizing a quarterly report containing a main report, annexes, metadata and an audit-status note. State what must never be silently truncated.

**Expected answer:** The budget must reserve instructions, question, authoritative report sections, metadata, audit-status note and output. Units, periods, definitions, footnotes and provisional status cannot be silently separated from values.

**Common error:** Ignoring source authority, version or accountable approval.

Marking: award up to 4 points using the general framework.

## Exercise 5 Decoding

Compare greedy, temperature, top-k and top-p decoding. Recommend and justify settings for a controlled first draft of an official narrative.

**Expected answer:** Greedy is deterministic token selection; temperature reshapes probabilities; top-k limits candidates by count; top-p uses cumulative mass. A low-variability setting may support drafting, but source verification remains mandatory.

**Common error:** Conflating fluent language with verified truth.

Marking: award up to 4 points using the general framework.

## Exercise 6 Attention

Interpret the notebook attention matrix. State one valid conclusion and two conclusions the matrix cannot support.

**Expected answer:** A valid conclusion describes measured weighting in the defined toy calculation. It cannot prove causal reasoning, factual support or the explanation of a production model's final output.

**Common error:** Reporting a notebook result without interpreting its boundary.

Marking: award up to 4 points using the general framework.

## Exercise 7 Groundedness

Decompose a generated financial summary into claims and design a claim-to-source verification table.

**Expected answer:** Material claims should be separated into value, unit, period, entity and status, with an authorized source span, verification result, reviewer and disposition for each.

**Common error:** Treating a prompt as the only control.

Marking: award up to 4 points using the general framework.

## Exercise 8 Evaluation

Develop an evaluation plan that separates fluency, groundedness, correctness, robustness, safety and decision fitness.

**Expected answer:** Report each dimension separately, define representative and challenge cases, preserve references and rubrics, test relevant languages and subgroups, and state that results apply only to the tested configuration.

**Common error:** Ignoring source authority, version or accountable approval.

Marking: award up to 4 points using the general framework.

## Exercise 9 Risk controls

Identify privacy, prompt-injection and excessive-authority risks in a retrieval-and-tool application. Assign one preventive and one detective control to each.

**Expected answer:** Privacy: minimize and authorize inputs, then monitor disclosure. Prompt injection: isolate untrusted content, then monitor anomalous instructions or tool sequences. Excessive authority: enforce least privilege and approvals, then audit actions and exceptions.

**Common error:** Conflating fluent language with verified truth.

Marking: award up to 4 points using the general framework.

## Exercise 10 Evidence chain

Specify the complete SRAI evidence record required before a generated output influences a consequential decision.

**Expected answer:** Record request, purpose, user, authorized context and versions, model, prompt, retrieval, tools, decoding settings, raw output, claim checks, exceptions, reviewer, decision, time and retention rule.

**Common error:** Reporting a notebook result without interpreting its boundary.

Marking: award up to 4 points using the general framework.

## Exercise 11 Official statistics

Explain why a numerically correct agricultural statistic may still be misleading. Identify all metadata that must travel with the value.

**Expected answer:** A value becomes misleading when detached from definition, unit, denominator, geography, population, reference period, revision status, uncertainty or disclosure conditions.

**Common error:** Treating a prompt as the only control.

Marking: award up to 4 points using the general framework.

## Exercise 12 Capstone pathway

Describe how Lessons 1-30 could progressively support construction of a small educational GPT-style system. Distinguish it from fine-tuning an open model and from building a frontier model.

**Expected answer:** A defensible pathway introduces tokenization, embeddings, causal attention, decoder blocks, optimization, generation, evaluation and checkpointing, culminating in a small teaching model. Fine-tuning adapts an existing model; frontier training requires vastly greater data, compute, engineering and governance.

**Common error:** Ignoring source authority, version or accountable approval.

Marking: award up to 4 points using the general framework.

## Total and disposition

| Total | Suggested interpretation               |
|-------|----------------------------------------|
| 43-48 | Strong independent mastery             |
| 34-42 | Competent with limited gaps            |
| 24-33 | Developing; targeted revision required |
| 0-23  | Foundational remediation required      |